

100 Days of Azure DevOps

Day 1 Handout — Who holds the spatula, which city, which building

1 - Azure & DevOps Foundations · Personal lab only · Educational content · Not a sales pitch

Architecture A — shared responsibility (same app, different Azure product)

Layer	IaaS (VM)	PaaS (App Service)	SaaS
Azure type	Microsoft.Compute/virtualMachines	Microsoft.Web/sites, Microsoft.Web/sites/functions	Microsoft 365, GitHub, Azure DevOps
Application / data	You: app, disks, backups	You: app + App Settings / Connection Strings	Vendor: tenant data under their SLA
Runtime / middleware	You: IIS, nginx, .NET, Node, Java	Vendor: Kudu / platform stack you pick	Vendor
OS / patching	You: Windows/Linux Update, NSG, disks	Vendor patches the worker	Vendor
Network / identity	You: VNet, NIC, NSG, public IP	You: VNet integration, Private Endpoint, Entra ID	Vendor + your tenant admin
Failure blast	You reboot the VM; you own the 2am patch	Swap slot / scale out; platform recycles workers	Wait on vendor status + your admin settings
Lab example	Standard_B1s in rg-day01-lab	Free/F1 or B1 App Service plan — do not deploy yet	dev.azure.com personal org (Day 5)

Architecture B — geography map (pick a region like a latency + residency decision)

Concept	Azure fact	Lab default
Geography	Compliance boundary (data residency)	India / nearest allowed geography
Region	Group of datacenters; ARM location key e.g. centralindia	centralindia (or nearest)
Paired region	CAF: some regions pair for Azure-initiated GRB	Note pair; do not dual-region a lab
Availability Zone	Independent datacenter in the region (zone 1/2/3)	Skip AZ SKUs for Day 1 — extra cost
Availability Set	Fault + update domains inside one region (classic VM HA)	Not needed for empty RG labs
Resource group	Lifecycle + RBAC + cost scope, not a security boundary by itself	rg-day01-lab
Subscription	Billing + ARM quota + RBAC root for most labs	Personal sub ID from portal.azure.com

Architecture C — pitfalls that burn money or pin you to the wrong city

Pitfall	What you see	Fix
Work tenant by habit	Directory picker shows employer Entra ID	Switch directory → personal Microsoft account only
No budget	Invoice surprise after a VM left running	Cost Management → Budgets → Rs 500 / \$20 email alert
Wrong region forever	ARM templates hard-code eastus; users in India wait	Set location once; reuse in az group create -l
IaaS for a website	You patch IIS on a VM for a hello-world	PaaS App Service / Functions unless you need the OS
AZ as a personality trait	Zone-redundant SKUs on an empty lab	Single-region RG; AZ later when SLA requires it
Orphan RG	rg-day01-lab sits empty for 90 days	Delete if unused, or reuse as Day 2 target

One-liner to remember

IaaS is your kitchen, PaaS is a shared kitchen, SaaS is takeout — region is the city, zone is the building.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal Microsoft account and personal Azure subscription only. No work SSO, no employer tenant.

- 1 portal.azure.com → sign in with the personal Microsoft account. Top-right directory picker: confirm it is NOT a work tenant.
- 2 Subscriptions → open the personal subscription → copy Name + Subscription ID into notes (never a work sub).
- 3 Cost Management + Billing → Budgets → Add → amount Rs 500 or \$20 → alert email = you. Save.
- 4 Resource groups → Create → Name rg-day01-lab → Region Central India (or nearest) → Review + create.
- 5 Open Cloud Shell (Bash) → az account show --output table → confirm the same subscription ID.
- 6 Browse Create a resource → Storage account or App Service create blade. Read SKUs. Do NOT deploy.
- 7 Optional: az group list --output table. Keep rg-day01-lab for Day 2 or delete if you created nothing inside.

Prove the personal subscription, then create the Day 1 RG

```
az account show --output table
az account show --query "{name:name, id:id, tenantId:tenantId}" -o jsonc
az group create --name rg-day01-lab --location centralindia
az group list --output table
```

Done checklist

- Can explain IaaS vs PaaS vs SaaS with one Azure resource type each
- Can explain Region vs Availability Zone without saying 'datacenter somewhere'
- Budget alert exists on the personal subscription
- rg-day01-lab exists in the chosen region (or you can recreate it)

Tomorrow — Day 2

Azure Portal, CLI & PowerShell basics